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A microfoundations approach to studying innovation in multinational subsidiaries

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Research Summary: We study antecedents of innovation performance for the subsidiaries of multinational enterprises (MNEs) using the microfoundations approach. Based on the upper echelon perspective, we argue that managers' characteristics, such as prior MNE work experience and industry experience, affect subsidiary innovation. We tested our hypotheses on a sample of 228 MNE subsidiaries from 11 countries. Results indicate that managers' industry experience serves as an external boundary-spanning capability and, therefore, it has a greater effect on autonomous subsidiaries. In contrast, managers' prior MNE work experience functions as an internal boundary-spanning capability and, therefore, it has a smaller effect on subsidiaries that are less autonomous or engage in R&D.

Managerial Summary: The success of an MNE now increasingly depends on its ability to generate knowledge anywhere in the world. Thus, the ability of foreign subsidiaries to generate innovation plays an increasingly important role in enhancing the performance of MNEs. In this regard, what factors determine the innovativeness of foreign subsidiaries is an important question for managers. Our study suggests that the international experience of the top management team (TMT) of a subsidiary and its CEO's industry experience positively affect subsidiary innovation. Furthermore, the TMT's international experience has a greater effect on the innovativeness of subsidiaries that remain dependent on their headquarters for knowledge transfer. In contrast, the CEO's industry experience has a greater effect on the innovativeness of autonomous subsidiaries.

KEYWORDS
autonomy, boundary spanning, managerial experience, MNEs, subsidiary innovation

1 | INTRODUCTION

The success of a multinational enterprise (MNE) now increasingly depends on its ability to generate knowledge anywhere in the world. Foreign subsidiaries have become an important source of new knowledge creation for MNEs; subsidiaries are developing innovations that are used by other subsidiaries and the MNE itself (Birkinshaw & Pedersen, 2008). As a result, the role of MNE subsidiaries has changed from that of passive recipients of knowledge to that of proactive generators of knowledge. There has been significant research on the innovation practices of subsidiaries in the MNE network (see review by Michailova & Mustaffa, 2012). However, the focus in much of this research has been on macro-level factors such as the role of MNEs and their relationships with subsidiaries. This research has not paid adequate attention to the importance of subsidiary managers in affecting subsidiary innovation. In this article, we redress this oversight.

In an exhaustive review of the literature on subsidiary knowledge flows, Michailova and Mustaffa (2012) identified factors such as characteristics of actors (MNEs and subsidiaries), characteristics of knowledge, and characteristics of relationships between actors in the context of innovation and knowledge flows in MNEs. Knowledge generation at its most basic level is a result of cross-border connections and interactions between senior managers in headquarters (HQ) and subsidiaries (Song, 2014). Given the importance of individual actors, Cano-Kollman, Cantwell, Hannigan, Mudambi, and Song (2016) called for incorporating the microfoundations perspective in international business (IB) research. Felin, Foss, and Ployhart (2015) identified the upper echelon theory as a suitable lens to study innovation from the microfoundations perspective because of its focus on the top managers in an organization. In the foreign subsidiary literature, scholars have acknowledged the role of expatriate managers in facilitating innovation at the subsidiary level (Choi & Johanson, 2012; Fang, Jiang, Makino, & Beamish, 2010). Clearly there is a linkage between managerial characteristics and innovation in foreign subsidiaries (Bantel & Jackson, 1989). However, this linkage has not been systematically examined in the context of foreign subsidiaries. The purpose of our study is to investigate the importance of subsidiary managers in subsidiary innovation. We develop our theoretical arguments by integrating the literature on upper echelon theory, boundary spanning in subsidiaries, and knowledge management and testing them on a sample of 228 foreign subsidiaries in 11 countries.

Our article makes important contributions to the IB and knowledge management literature. First, using the microfoundations approach, we argue that subsidiary managers act as boundary spanners between the parent firm and its subsidiaries. In this role, subsidiary managers can help integrate and diffuse knowledge across MNC networks throughout the innovation process. In doing so, we contribute to the emerging literature on the role of boundary spanners in global organizations (Schotter, Mudambi, Doz, & Gaur, 2017). Second, much of the extant literature measures innovation using proxies such as research intensity, number of patents, and patent citations; these measures have limitations because they reflect inputs and the intermediate stage in the innovation process, respectively.

Instead, we rely on new product development as a measure of subsidiary innovation, which is a more accurate, outcome-based measure of innovation. Finally, we argue that subsidiary-level factors such as autonomy and technological capabilities are important contingencies that affect the boundary-spanning capabilities of subsidiary managers. Specifically, we assert that subsidiary managers' prior MNC work experience is an internal boundary-spanning capability and is less important in subsidiaries that are autonomous or engage in research and development (R&D). In contrast, the industry experience of the top manager of a subsidiary is an external boundary-spanning capability that is more important in subsidiaries that are autonomous or engage in R&D.

2 | THEORY AND HYPOTHESES

MNE success in international expansion has been attributed to MNEs' unique ability to generate, transfer, and exploit knowledge in their internal network of differentiated subsidiaries (Gupta & Govindarajan, 1991, 2000; Kogut & Zander, 1993). Literature in this domain has highlighted the role of subsidiary-driven initiatives (Birkinshaw, 1997; Birkinshaw & Morrison, 1995), the importance of subsidiaries to the MNE (Andersson, Forsgren, & Holm, 2007), subsidiary mandates in R&D (Cantwell & Mudambi, 2005; Scalera, Mukherjee, Perri, & Mudambi, 2014), and knowledge flows and power relations between HQ and subsidiaries (Mudambi & Navarra, 2004) as determinants of innovation performance.

Foreign subsidiaries are themselves seen as creators of new knowledge (Cantwell & Mudambi, 2005; Frost, 2001). Un (2011) proposed that foreign subsidiaries are successful at innovation because they are subsidized by the MNE network and can leverage their innovation costs across the network. Furthermore, foreign subsidiaries' innovativeness can be partially explained by their internal embeddedness (Andersson, Dasi, Mudambi, & Pedersen, 2016). Internal embeddedness enables a foreign subsidiary to take advantage of other subsidiaries and HQ for access to resources, support, and exchanges (Mukherjee, Lahiri, Ash, & Gaur, 2017); these help subsidiaries leverage their own innovation effects (Un, 2011). Subsidiary managers play a critical role in helping establish the linkages and pipelines between the focal subsidiary and other parts of the MNC network (Gaur, Delios, & Singh, 2007). Subsidiary managers, thus, act as boundary spanners, enabling subsidiaries to pursue innovation (Kostova, Marano, & Tallman, 2016).

The HQ–subsidiary relationship exists in a complex network—one that involves connectivity at several levels. Andersson et al. (2016) identified two levels of connectivity: (a) organizational-based pipelines created and maintained by MNEs and (b) individual-based personal relationships within this network. They further stated that the vast majority of connectivity research has focused on the organizational level and very little at the individual level. Knowledge generation and transfer has a tacit aspect (Nonaka, 1994); tacit knowledge can be disseminated in the MNE only through experienced managers (Fang et al., 2010; Tan & Mahoney, 2006). Given the importance of individual actors in the knowledge management processes, scholars have suggested applying the microfoundations approach to study the genesis of knowledge management in the MNE network (Foss & Pedersen, 2004). Felin et al. (2015) defined microfoundations as concerned with understanding how actors, their interactions, and the mechanisms and context that influence such interactions produce firm-level and collective heterogeneity. The microfoundational perspective, thus, provides a big tent for multiple subfields in strategic management that use lower-level constructs to explain higher-level organizational outcomes.

Within this broad operational definition, the upper echelon perspective can be viewed as microfoundational in nature because "it explains how characteristics, role, and socio-psychological dynamics of top management influence organizational-level outcome" (Felin et al., 2015, p. 585). Compared to the broader movement of the microfoundational approach, however, the upper echelon perspective focuses only on a handful of high-ranking individuals in the organization. According to this view, knowledge resides in organizational actors and is influenced by organizational structure (e.g., hierarchy) that shapes the interactions among different actors (Grant, 1996). Nevertheless, more work is needed to link the upper echelon perspective to the broader microfoundations approach. Recent studies on boundary-spanning capability (Schotter et al., 2017) do provide the link between the upper echelon perspective and the broad definition of microfoundations; this literature incorporates the social interaction between key individuals within the organization as well as with the stakeholders in the external environment.

Certain organizational actors may have superior knowledge as well as boundary-spanning capabilities because of their unique experiences or position in the MNC network. In a microfoundations-based study of firm innovation, Grigoriou and Rothaermel (2014) found that the presence of relational stars (integrators and connectors) had a positive effect on firm knowledge advantage. Similarly, Liu, Wright, Filatotchev, Dai, and Lu (2010) found that new ventures founded by individuals with foreign experience tend to be more innovative than those founded by individuals without foreign experience. Maitland and Sammartino (2015) developed a cognition-based perspective of firm internationalization, in which they proposed that executives with international experience are richer and more connected in their mental processes than executives without international experience. In the next section, we focus on how specific types of experiences of subsidiary managers are linked to innovation outcomes at the subsidiary level. We conduct a systematic review of the extant literature (Gaur & Kumar, 2017) and develop our theoretical model relying on the upper echelon theory (Hambrick & Mason, 1984) and the boundary-spanning literature (Schotter et al., 2017), which have been identified as relevant lenses to capture the links between managerial characteristics and organizational outcomes such as innovation (Felin et al., 2015).

2.1 | Hypotheses

The upper echelon perspective suggests that top managers in a firm affect the strategic choices and performance outcomes at the firm level. Hambrick and Mason (1984), for example, stated that demographic factors, functional expertise, and career experiences of the top management team (TMT) members affect firm performance. The underlying premise is that executives are influenced by their backgrounds and, thus, develop expectations, preferences, mind-sets, values, and behaviors based on experiences they encounter. Managers are also driven by their belief systems in understanding their external environments (Perlmutter, 1969). These belief systems, in turn, are influenced by factors such as education, age, functional background, and experience (Prahalad & Bettis, 1986).

In the IB field, scholars have utilized the upper echelon perspective to examine the linkage between TMT experience and firm internationalization. Sambharya (1996) found that the foreign experience of the TMT members was related to their firm's international diversification strategies. Tihanyi, Ellstrand, Daily, and Dalton (2000) found that TMT average tenure, education, international experience, and tenure heterogeneity were positively linked with global strategic posture. Likewise, Reuber and Fischer (1997) found that internationally experienced top management teams have a greater propensity to develop foreign business relations which, in turn, help the firm gain competitive advantage in international markets.

A similar perspective can also be found in the knowledge management literature. Grant's (1996) knowledge-based theory proposed a microfoundation-based mechanism of firm knowledge capability; he emphasized the role of individuals, such as inventors and managers, in the knowledge creation process. Scholars have shown that TMT characteristics are associated with innovation outcomes (Bantel & Jackson, 1989; Simsek, 2007). For example, Simsek (2007) found that top managers affect innovation outcomes because of their experience and superior knowledge of the external environment.

A key mechanism through which subsidiary managers affect innovation is their role as boundary spanners. Schotter et al. (2017) defined boundary spanning in global organizations "as a set of communication and coordination activities performed by individuals within an organization and between organizations to integrate activities across multiple cultural, institutional and organizational contexts" (p. 404). Foreign subsidiaries are often embedded in the host market contexts as well as in the network of other subsidiaries of the same MNC parent (Gaur & Lu, 2007). To derive benefits from this dual embeddedness, subsidiary managers must work as conduits of information flow, as well as negotiators with the external stakeholders, including the MNC parent. Thus, subsidiary managers are nested antecedents to subsidiary-level outcomes, including innovation (Barney & Felin, 2013; Felin, Foss, Heimeriks, & Madsen, 2012).

In the following sections, we develop our arguments linking subsidiary managerial characteristics to subsidiary innovation by looking at the MNC and industry experience of the subsidiary TMT members. We argue that MNE experience represents an internal boundary-spanning capability, whereas industry experience represents an external boundary-spanning capability (Andersson et al., 2016; Cano-Kollman et al., 2016). In addition, we examine how the subsidiary autonomy and its focus on R&D conditions the relationship between managerial experience and subsidiary innovation.

2.2 | MNE experience and subsidiary innovation

We propose that MNE experience is a surrogate for the expatriate experience. Managers with MNE experience develop multiple perspectives because of exposure to several organizational contexts and environments. Scholars have argued that experiential learning could be critical for the job performance of managers (Sambharya, 1996). Experiential learning also helps subsidiary managers in undertaking the boundary-spanning functions. With greater experience, managers develop a greater understanding of the organizational dynamics within the MNE network and greater leverage against different stakeholders, including the senior managers from the MNE parent. This enables subsidiary managers to effectively negotiate on behalf of the subsidiary and work as technology scouts to obtain novel knowledge from external sources and bring it to the focal subsidiary (Klueter & Monteiro, 2017; Schotter et al., 2017).

The knowledge management literature suggests that the knowledge needed for innovation can be obtained from a variety of internal and external sources. A study by Mansfield (1988) found that original sources of invention came from outside the firm. This was further supported by Jaffe, Trajtenberg, and Fogarty (2000), who found that knowledge created by one party creates positive externalities by stimulating the innovation capabilities of other parties. However, much organizational knowledge is tacit in nature (Nonaka & Takeuchi, 1995) and socially complex, which makes knowledge transfer very difficult (Contractor, Yong, & Gaur, 2016; Kogut & Zander, 1993; Teece, 1979). Human mobility, such as mobility from one organization to another organization, can expedite the interorganizational knowledge transfer effectively because it brings the tacit knowledge attached to the person from one organization to another (Gaur, Kumar, & Singh, 2014; Song, Almeida, & Wu, 2003). Furthermore, the boundary-spanning literature argues that the top managers work as conduits of the knowledge they gain from past work experience in other MNEs. Such knowledge and competence help subsidiary managers in opportunity formation which, in turn, helps in the development of creative solutions for the focal subsidiary (Tippmann, Sharkey Scott, & Parker, 2017).

Fang et al. (2010) identified two factors that characterize the pivotal role of subsidiary managers as boundary spanners in the transfer of knowledge in MNEs: (a) increasing formal and informal interunit communications between headquarters and their subsidiaries and (b) increasing interunit homophily (Gupta & Govindrajan, 2000). Managers with HQ experience are well positioned to improve the scope and richness of knowledge transfer through both formal and informal channels (Gupta & Govindrajan, 2000). Experienced managers can take advantage of their connections in the parent company to obtain knowledge regarding strategic and operational issues and practices that are useful to subsidiaries. They also have extensive informal ties in the parent company and other foreign subsidiaries because of the socialization, personal contacts, and interpersonal familiarity (Gaur et al., 2007) that further facilitate knowledge transfer and add to the subsidiary's absorptive capacity (Fang et al., 2010).

Boundary-spanning managers are, thus, crucial to enhance the subsidiary's absorptive capacity by increasing the interunit homophily between the subsidiary and its parent (Fang et al., 2010; Schotter et al., 2017). Absorptive capacity is the degree to which new information is recognized, assimilated, and applied to commercial ends (Cohen & Levinthal, 1990), and it is critical to the knowledge transfer between the parent and its subsidiary (Datta, Mukherjee, & Jessup, 2015; Lane, Salk, & Lyles, 2001). Absorptive capacity is related to the level of homophily—"the degree to which two or more individuals who interact are similar in certain attributes such as beliefs, education, social status, and the like" (Rogers, 1995, pp. 18–19). The presence of managers with MNE experience in a local subsidiary would increase the level of interunit homophily between the parent and the subsidiary (Fang et al., 2010). In a microfoundation-based study of the role of a certain individual in firm innovation, Grigoriou and Rothaermel (2014) found that the presence of integrators and connectors had a positive effect on firm knowledge advantage. They defined integrators as managers who have an extraordinarily large, extensive, and dense network of intrafirm collaborators and experience in a large MNE.

These arguments find support in prior studies of upper echelons that suggest a positive relationship between managers' prior international experience and organizational learning. There are at least three mechanisms through which managers with greater MNE experience facilitate innovation at the subsidiary level. First, past work experience for different MNEs facilitates the accumulation of culturally diverse knowledge (Singh, Gaur, & Schmid, 2010). This cross-cultural competence can enhance managers' abilities to work with people from different cultural backgrounds. In addition, working in a cross-cultural environment enhances foreign language ability, which can improve the communication between the subsidiary managers and the headquarters representatives. Communication skills can help ease coordination between the subsidiary and the headquarters, thereby fostering knowledge transfer.

Second, international exposures familiarize managers with the routines of multinational operations. According to Black, Gregersen, and Mendenhall (1992), managers of global firms encounter three strategic functions: (a) development and succession of the management team, (b) coordination and control of global operations, and (c) information exchange between parent and subsidiaries and among subsidiaries. Familiarity with these three strategic functions, as well as the rigid bureaucracy of MNEs, enables managers to better interact with headquarters. Consequently, subsidiary managers with MNE experience can better facilitate resource transfer, including knowledge, from headquarters to subsidiaries.

Third, prior exposure to global business can shape managers' cognition by enriching informational diversity (Mihalache, Jansen, Van Den Bosch, & Volberda, 2012), overcoming home-country bias and developing a global mind-set (Perlmutter, 1969; Prahalad & Bettis, 1986). Managers with more international exposure are more conversant in foreign ideas and, therefore, are more likely to absorb them than managers with less international exposure. This ability, in turn, helps managers combine knowledge from various sources and enhance the innovation capability of their subsidiaries. Consistent with these arguments, Haas (2006) found that cosmopolitan members of transnational teams had more technical knowledge and more internal and external knowledge than local members, at both the individual and team levels. In summary, we argue that managers' international exposure gained through previous MNE experience is positively associated with subsidiary innovation.

> **Hypothesis 1 (H1)** *Subsidiaries that have a higher proportion of managers with MNE experience are more likely to be innovative.*

2.3 | Industry experience and subsidiary innovation

Managers' industry experience can be viewed as an organizational resource as well as tacit knowledge that can help firms gain competitive advantage in the marketplace. In an early study, Gupta (1984) argued that the longer the managers' experience was in one particular industry, the more familiar they become with the industry structure and competitive strategy. At the level of strategic business units (SBU), Gupta asserted that familiarity with one particular industry increases executives' understanding of consumer needs and competitors' behavior which, in turn, provides valuable information for an SBU's strategic decision making, especially if the SBU has a mission to create competence rather than simply exploit the competence of its respective business group.

In the context of an MNE global operation, we argue that subsidiary managers' industry experience reflects the external boundary-spanning capabilities of its managers and thereby influences subsidiary innovation. There are several mechanisms for this effect. First, following Gupta (1984), we assert that managers with longer experience in a specific industry are more likely to be familiar with the industry structure and the potential strategies of firms in that industry. The more time one spends working in an industry, the better his/her ability will be to observe and collect sufficient information on the behavior of rival organizations in the same industry. This information and experience expedite the diffusion of innovation in a particular industry. This notion is also consistent with the study of executive migration and strategy diffusion (Rogers & Shoemaker, 1971). Rogers and Shoemaker (1971) found that the recruitment of executives from competitors in the same industry increases the likelihood of firms to promote or adapt a competitor's strategy because the newly recruited executives provide firsthand experience of executing the adapted strategy. Boeker (1997) also found that executive migration from competitors in the same industry increases the likelihood of a firm to enter a new product market that has been entered earlier by its rival.

Second, longer careers in a particular industry enhance the accumulation of knowledge and awareness of the technological trends in the industry. Awareness of technological trends helps managers gather information on current as well as future technologies that may shape the future of the industry. Familiarity with potential technological changes makes managers more open to investments in innovation that are necessary to anticipate the changes in the future. Additionally, longer industry experience enhances a manager's ability to select innovation that suits the organizational needs as well as organizational capability to compete in the market. Thus, industry experience helps in opportunity identification, which is an important prerequisite for innovation in subsidiaries (Tippmann et al., 2017).

Third, experience in a given industry increases a manager's ability to exchange and combine knowledge from suppliers and customers. Subsidiary managers who have been through the ups and downs of the industry cycle are in a better position to accumulate sufficient information on market trends, which enables them to better design products that meet the needs of local customers. The knowledge of local suppliers and customers increases the knowledge bases of managers, and the combination of this knowledge, in turn, increases the subsidiary's learning capacity (Smith, Collins, & Clark, 2005). Managers with greater industry experience can also be viewed as connectors who can collaborate with previously unconnected external knowledge centers and recombine knowledge from distant untapped knowledge clusters (Grigoriou & Rothaermel, 2014).

Fourth, industry experience is hard for competitors to imitate because of imperfections in the managerial labor market (Kor, 2003). In inefficient labor markets, the supply of top managers is limited. Top managers with industry-specific experience can also reduce the liability of foreignness because of their knowledge of what can work in new markets. Kor (2003) found that industry-level experience contributed to team competence that led to future firm growth. Finally, managers with specific industry knowledge will also be more effective, even in the internal boundary-spanning activities and in knowledge transfer between headquarters and subsidiaries (Choi & Johanson, 2012). Thus, managers with greater industry experience work as effective boundary spanners, bridging the gap between the subsidiaries and their external environment and, thereby, facilitating subsidiary innovation.

There are some counterarguments about the effect of managers' industry experience on innovation. For example, Hambrick and Mason (1984) argued that top managers who have spent their entire careers in one firm or one industry are more likely to focus on current markets than new ones, as they develop inertia to change. Similarly, Bantel and Jackson (1989) suggested that same-industry experience is associated with limited information search and overreliance on restricted knowledge, which can cause bias toward the *status quo*. However, we maintain that greater industry-specific experience does not necessarily imply that managers reject all types of innovation. There is evidence that innovations that come through the current strategic frame are more likely to receive support than innovations that come from outside the current strategic frame (Burgelman, 1991). Likewise, Tushman and Anderson (1986) argued that competence-enhancing innovations are more likely to be introduced by industry incumbents, suggesting that industry experience of top managers is likely to be associated with industry-relevant innovation. Accordingly, we hypothesize the following:

> **Hypothesis 2 (H2)** *Subsidiaries that have top managers with greater industry experience are more likely to be innovative.*

In our hypotheses, we have argued the importance of subsidiary managers as individuals who help in spanning internal and external boundaries which, in turn, enhance subsidiary innovation. However, the relative importance of global integration and local adaptation may condition the effectiveness of subsidiary managers in enhancing subsidiary innovation. The trade-off between global integration and local adaptation determines the extent to which a subsidiary is capable of pursuing innovative activities. Prior studies on headquarters-subsidiary relationships (Bartlett & Ghoshal, 1986; Gupta & Govindarajan, 1991) have suggested that the relative importance of global integration and local adaptation is reflected in the degree of subsidiary autonomy, while the trade-off between parent and local knowledge is reflected in subsidiary R&D activity. Following this, we argue that subsidiary autonomy and R&D activity moderate the relationship between managers' experience and innovation.

2.4 | Subsidiary autonomy as a moderator

One of the important issues in MNE global operations is the decision about the hierarchical relationship and autonomy between the headquarters and its subsidiaries. A greater level of hierarchical control can ensure that subsidiaries conform to the pursuit of the global objective of the firm. The hierarchical relationship enforces the effectiveness of organizational fiat, which can ensure that the subsidiary managers have no incentive to deviate from its mandate (Brandenburger & Stuart, 2007). Asymmetric information between the headquarters and its subsidiaries creates opportunities for subsidiary managers to pursue their self-interests, which may not correspond to the objectives set by the headquarters (Lee & Gaur, 2013). Organizational fiat can reduce the probability of such opportunistic behavior by exerting greater monitoring and control (Andersson, Gaur, Mudambi, & Persson, 2015). Furthermore, as suggested by Williamson (1979), organizational fiat can also reduce hazards related to recurrent transactions. In the context of MNE global operations, such hazards potentially occur when a subsidiary represents an important node within an MNE's global value chain such that any production delay in the subsidiary may cause interruption in the global production. Hierarchical relationships, through fiat, can ensure that the probability of such hazards is low by allowing the headquarters to manage its subsidiaries in such a way that recurrent resource transfer between the headquarters and its subsidiaries runs efficiently and the production activity conforms to the standard requirement of global operation.

Moreover, hierarchical relationships streamline the coordination between the headquarters and its subsidiaries. Strong coordination is especially important when there is a need to allocate resources from the headquarters to its subsidiaries. As suggested by Teece (1979), transfer of capability is costly and difficult, but strong coordination can ease these difficulties. Moreover, coordination is necessary when the MNE emphasizes global standardization more than local adaptation of its products. Global standardization can be important when an MNE focuses on global production to achieve efficiency through economies of scale or when the degree of competitive intensity in the local market is low.

Gupta and Govindarajan (1991) asserted that hierarchical governance is superior to autonomy when a subsidiary is the user, not the creator, of knowledge developed elsewhere; a hierarchy allows for strong coordination during the process of resource or knowledge transfer. Similarly, hierarchical governance is relatively superior when an MNE operates in industries in which the home country has a comparative advantage over the host country. In such industries, the home country's comparative advantage, often in the form of tacit capabilities, must be passed on to a subsidiary so that it enables the subsidiary to overcome the liability of foreignness in the host country.

In the presence of greater hierarchy (less autonomy), subsidiary managers must traverse the boundaries between the subsidiary and the headquarters. Such boundary spanning helps in securing resources from the headquarters and developing innovation capabilities at the subsidiary level (Schotter et al., 2017). The role of subsidiary managers as integrators may be diminished to some extent in any situation where the subsidiary is strong and has greater autonomy. Additionally, it may be important for the subsidiary managers to get support from the headquarters to change the subsidiary charter such that it is allowed to indulge in local innovation (Cantwell & Mudambi, 2005). Consistent with our previous argument on international experience, subsidiary managers with prior MNE work experience can better facilitate such information exchange and expedite the transfer of capability or resources from headquarters to the subsidiary. Thus, MNE experience enables subsidiary managers to become more effective internal boundary spanners and negotiate the charter of the subsidiary with headquarters.

In contrast, when a subsidiary has a greater level of autonomy, the need to negotiate the subsidiary charter or to transfer resources and capabilities is much less. While the MNE experience may still be helpful, we argue that it has less value in subsidiaries with a high level of autonomy than in subsidiaries with a low level of autonomy. Accordingly, we hypothesize that the headquarters–subsidiary relationship conditions the relationship between managers' international experience and subsidiary innovation.

> **Hypothesis 3a (H3a)** *Subsidiary autonomy negatively moderates the relationship between the TMT MNE experience and subsidiary innovation such that the positive effect of the TMT MNE experience on subsidiary innovation is weaker when the subsidiary is autonomous.*

The boundary-spanning role of subsidiary managers is not limited to internal organizational boundaries. Managers often must also traverse the external boundaries in the local environment to understand the expectations of the local market and gain critical location-specific resources and capabilities (Gaur et al., 2007). The need for such external boundary spanning and connection is greater in subsidiaries that have greater levels of autonomy.

In contrast to hierarchy, autonomy embraces local knowledge and encourages subsidiaries to make decisions or choose strategies that suit local markets. According to Gupta and Govindarajan (1991), subsidiary autonomy is critical when the subsidiary is a creator rather than a user of MNE global knowledge. Knowledge creation requires the subsidiary to absorb knowledge, capabilities, and resources embedded in the host country; this capability to absorb is maximized when the subsidiary has decision-making authority. In the context of MNEs from emerging markets, Wang, Luo, Lu, Sun, and Maksimov (2014) argued that autonomy at the subsidiary level can help emerging market MNEs overcome the latecomer disadvantage by allowing their subsidiaries to independently tap into strategic resources in the host countries. Moreover, decision-making authority can enable subsidiaries to develop new products or to introduce processes or organizational innovation locally with limited dependencies on the headquarters. This decision-making ability is especially important when the products require a high degree of local adaptation. Under such circumstances, subsidiary autonomy allows managers to quickly adapt the products based on local market needs.

Subsidiary managers with a greater level of industry experience are more effective boundary spanners and connectors of the local boundaries that must be traversed in an autonomous subsidiary. Subsidiary autonomy lends power and authority to experienced managers to exchange and combine knowledge with local suppliers and competitors and work as technology scouts (Klueter & Monteiro, 2017; Schotter et al., 2017). In addition, subsidiary autonomy allows managers with sufficient industry experience to execute innovation strategy based on their professional judgment in the trajectory of the local market and competition in the future. Hence, we argue that the positive effect of managers' industry experience on subsidiary innovation is stronger when there is a greater degree of subsidiary autonomy.

> **Hypothesis 3b (H3b)** *Subsidiary autonomy positively moderates the relationship between the top manager's industry experience and subsidiary innovation such that the effect of the industry experience on subsidiary innovation is stronger when the subsidiary is autonomous.*

2.5 | Subsidiary R&D as a moderator

While in the past MNE headquarters was often the source of knowledge and innovation for subsidiaries, in recent years, with the globalization of innovation activities around the world, many subsidiaries have started to play an active role in knowledge creation (Cantwell & Mudambi, 2005). The increasing R&D intensity at the subsidiary level attests to this change in the charters of subsidiaries around the world, including in developing countries (Mudambi, 2011). Our baseline argument is that a higher level of subsidiary R&D is associated with a greater level of subsidiary innovation. We elaborate the mechanism for this direct effect, before discussing the moderating effects.

MNEs offshore their R&D facilities to foreign subsidiaries for various reasons. In the past, the main purpose for R&D offshoring was to adapt the products to the local markets or to adjust production process to fit local resources (Cantwell & Mudambi, 2005). In this model, subsidiary R&D can be seen as the subsidiary's absorptive capacity to further improve the utilization of knowledge from the home country. However, as innovation is becoming increasingly global and firms and new technology are clustered in many different countries, foreign subsidiaries have started to take a more active role in knowledge creation. Foreign research facilities enable MNEs to transcend limitations in the home country's technological specialization and pursue different technological sets abroad (Cantwell, 1992, 1993).

In addition, the technology clusters in various geographical regions further promote technology specialization across countries. According to data from the National Science Foundation (2012), for example, foreign countries have outpaced the patenting activity of the United States, except in the fields of information and communication technologies, automation, biotechnology, and pharmaceuticals, where the United States remains the biggest player. The declining dominance of the United States as a knowledge creator motivates MNEs to strategize the role of their foreign subsidiaries to take advantage of host country knowledge (Berry, 2015). For this reason, we can argue that subsidiary R&D is part of MNE strategy to explore local knowledge, especially when the host country has a comparative advantage in a particular field of technology. Our argument is supported by Frost (2001), who found that subsidiaries are more likely to rely on local innovation (or local patents) when the host country has a comparative advantage in the particular technical field of that patent. Thus, subsidiary R&D is an effort by MNEs to secure and integrate local knowledge for basic innovation.

As we argued before, MNE experience represents the internal boundary-spanning capability, whereas industry experience represents the external boundary-spanning capability of the subsidiary managers. The internal boundary-spanning capability is likely to be more important when subsidiaries must traverse internal boundaries to secure access to resources from the headquarters. Such internal boundary spanning becomes more important when subsidiaries do not engage much in R&D themselves. In such cases, managers may bring the information about innovations and new product developments from elsewhere in the MNE network, which would enable the focal subsidiary to introduce new products in the local markets. In contrast, when subsidiaries engage in more local R&D, the importance of internal boundary-spanning capabilities and the need for integrators drops. Accordingly, we argue that the effect of international experience on subsidiary innovation is less pronounced when subsidiaries are engaged in R&D.

In contrast, the external boundary-spanning capability, as reflected by the industry experience of subsidiary managers, is more important when subsidiaries engage in local R&D. The underlying mechanism for this relationship is that managers' industry experience increases a subsidiary's responsiveness to changes in the local environment, as well as the subsidiary's capabilities to absorb information from its local competitors, suppliers, and consumers. A subsidiary's R&D activity extends subsidiary knowledge search capabilities (Un, 2011) and shortens the time to search for local knowledge. The subsidiary's R&D, therefore, augments the proposed relationship between managers' industry experience and innovation by facilitating the creation or combination of knowledge that can enhance the subsidiary's absorptive capacities. These two contrasting effects of internal and external boundary-spanning capabilities on the relationship between subsidiary R&D and innovation lead to the following hypotheses:

Hypothesis 4a (H4a) *A subsidiary's R&D negatively moderates the relationship between the top manager's MNE experience and subsidiary innovation such that the positive effect of MNE experience on subsidiary innovation is weaker when the subsidiary performs R&D activity.*

Hypothesis 4b (H4b) *A subsidiary's R&D positively moderates the relationship between the top manager's industry experience and subsidiary innovation such that the effect of industry experience on subsidiary innovation is stronger when the subsidiary performs R&D activity.*

Figure 1 summarizes the theoretical model proposed above.

3 | METHODS

3.1 | Sample and data

We draw our data from the Management, Organization and Innovation (MOI) Surveys, administered jointly by the European Bank for Reconstruction and Development (EBRD) and the World Bank (Bloom & Van Reenen, 2007). The purpose of these surveys is primarily to collect data on management practices across countries (Bloom, Schweiger, & Van Reenen, 2011). EBRD and the World Bank conducted face-to-face interviews in 12 countries, where interviewers were recruited by local survey companies. The 12 countries included in the surveys are Belarus, Bulgaria, Germany, India, Kazakhstan, Lithuania, Poland, Romania, Russia, Serbia, Ukraine, and Uzbekistan.

Firms included in the survey are in the manufacturing industry and, thus, the survey primarily targets factory, production, and operation managers. However, the respondents often have a senior position in the firm, such as chief executive officer (CEO), vice president, or general manager.

Our dataset in this study is drawn from 11 countries. Because of the small number of observations of foreign subsidiaries located in Belarus (only four), we do not include them in our study. We collected data on 1,675 responding firms from the remaining 11 countries. However, because we are interested only in foreign subsidiaries, we choose firms under the category of foreign ownership. With these criteria, we have 228 subsidiaries from 11 countries in our dataset. Table 1 presents the distribution of MOI's responding firms in each country and the number of foreign-owned subsidiaries in each country, as well as their proportion in MOI surveys. For comparison, we present the

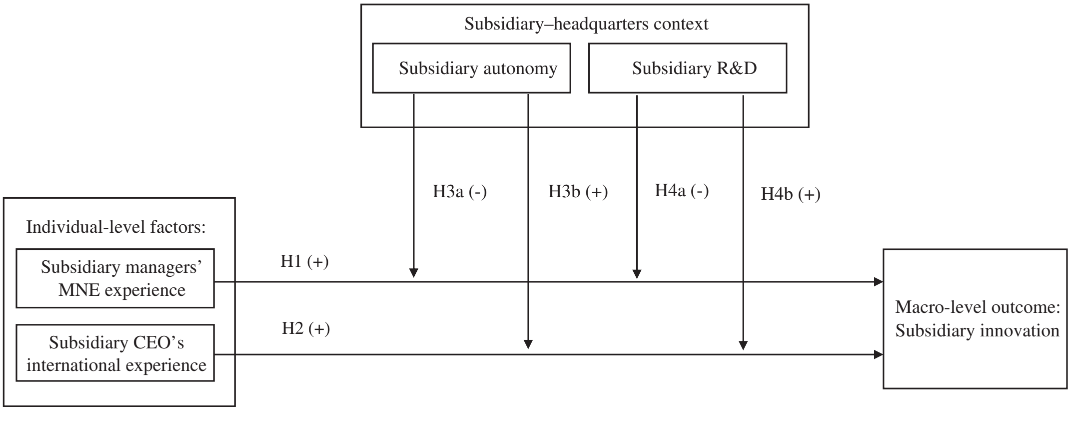

FIGURE 1 The hypothesized model

shares of foreign-owned MNEs based on Bureau van Dijk's Orbis database (Bloom et al., 2011). According to Bloom et al. (2011), the sampling proportion (based on ownership and size) in MOI surveys attempts to mirror the structure in Bureau van Dijk's Orbis database.

3.2 | Dependent variable

We operationalize our dependent variable as product innovation and measure it using a binary variable. This variable takes the value of "1" if a subsidiary introduces a new product or significantly modifies an existing product within the last 3 years and "0" otherwise. This construct operationalization is similar to the construct used in Un (2011, 2016), who measured firm innovation through the introduction of new products.

3.3 | Explanatory variables

There are two main explanatory variables in this study: subsidiary managers' prior MNE work experience and their industry experience. We measure prior MNE work experience by the percentage of subsidiary top managers with work experience in different MNCs prior to joining the focal firm. The top manager's industry experience is measured by the number of years the top manager has worked in the same industry as that of the subsidiary.

There are two moderating variables: subsidiary autonomy and subsidiary R&D. Autonomy is defined as the ownership of both financial authority and task accountability (Vancil, 1979). Following this idea, an autonomous unit must have both financial accountability and decision-making authority. For this reason, Weigelt and Miller (2013) measured a business unit's autonomy as a dichotomous construct and not as a continuous construct. Adopting Weigelt and Miller's (2013) definition, we measure foreign subsidiary autonomy by taking into consideration the subsidiary's financial accountability and decision-making authority in four respects: new product introduction, hiring, product pricing, and advertising authority. Specifically, we construct subsidiary autonomy as a binary variable that takes a value of "1" if the subsidiary (a) has a separate financial statement from headquarters, (b) makes hiring decisions, (c) makes decisions on new product introduction, (d) makes product pricing decisions, or (e) makes advertising decisions; it is "0" if the subsidiary does not have control of any one of the aforementioned aspects.

TABLE 1 Number of observations per country

Country	Total firms in MOI surveys	Foreign-owned firms (% share)	Share of foreign-owned firms on BvD's Orbis database
Bulgaria	154	26 (16.9%)	4.6%
Germany	222	39 (17.6%)	15.8%
India	200	8 (4.0%)	4.0%
Kazakhstan	125	6 (4.8%)	2.4%
Lithuania	100	19 (19.0%)	13.0%
Poland	103	22 (21.4%)	13.7%
Romania	152	41 (26.9%)	2.6%
Russia	214	8 (6.5%)	0.97%
Serbia	135	22 (16.3%)	7.4%
Ukraine	147	7 (4.8%)	0.0%
Uzbekistan	123	30 (24.4%)	1.6%

Source: MOI surveys and Bloom et al. (2011).

3.4 | Control variables

We control for subsidiary-level factors as well as host country-level factors. Subsidiary-level factors include subsidiary size, subsidiary age, and an indicator variable to identify whether or not a subsidiary registered a patent. A subsidiary's size is measured by the log of the number of its employees. We also control for the age of the subsidiary's CEO and the average education of the subsidiary's managers. In addition, we control for unobserved industry factors by creating a binary variable that equals "1" if a subsidiary belongs to a high-tech industry and "0" otherwise. We identify high-tech industries based on the definition set by the Bureau of Labor Statistics (National Science Foundation, 2012).

Host country-level characteristics include property rights protection, which is measured by the International Property Rights Index (IPRI), provided by Property Rights Alliances (2015). The index takes a value from "1" (low protection) to "7" (high protection). This index assesses the extent to which economic activity is facilitated by an effective legal and political environment, protection of physical property rights, and protection of intellectual property rights. We take the 3-year average value of IPRI scores. We also control for the size of the city (in terms of population) where the subsidiary is located by constructing ordinal variables of −5 (1 = population less than 50,000, 2 = population between 50,000 and 250,000, 3 = population over 250,000 to 1 million, 4 = city other than the capital with a population over million, and 5 = capital city).

3.5 | Econometric model

Because of the categorical nature of our dependent variable, we use binomial logit estimation. Logit estimation is particularly useful to deal with binary dependent variables because it avoids the unboundedness problem (estimated value falls below 0 or above 1) found in linear probability models. In addition, we acknowledge the possibility of unobserved host country characteristics that may affect the accuracy of our estimation. We deal with this problem by employing clustered standard errors with host countries as the origin for clustering by assuming that host country unobserved characteristics are the source of correlation across subsidiaries operating in that country.

4 | RESULTS

4.1 | Descriptive statistics

Table 2 displays the descriptive statistics for our variables. Sixty-nine percent (157 out of 228) of subsidiaries in our sample introduced new products within the last 3 years. Additionally, 58% of subsidiaries are autonomous and 45% perform R&D. The average proportion of managers with MNE prior work experience is 16.30%, while the mean of the top manager's industry experience is 14.67 years.

Table 3 displays the correlation matrix for all variables. None of the variable pairs has a correlation coefficient above 0.3, indicating that multicollinearity is not a concern in our models.

We further examine the association between binary variables in our model (subsidiary autonomy, R&D, subsidiary patent, and product innovation) using cross-tabulation and Cramer's V. The results are displayed in Table 4. None of the variable pairs, especially pairs of independent variables, has a Cramer's V greater than 0.5. This again confirms that multicollinearity between binary independent variables does not affect our results.

TABLE 2 Descriptive statistics

	# Obs.	Mean	Std. dev.	Min.	Max.
Product innovation	228	0.69	0.46	0	1
Subsidiary size	228	5.23	0.99	3.43	8.59
Subsidiary age	228	3.00	0.89	0.69	5.14
Subsidiary patent	228	0.37	0.48	0	1
Size of subsidiary location	228	2.92	1.49	1	5
High-tech industries	228	0.17	0.38	0	1
Intellectual property rights protection	228	5.39	1.19	3.90	7.60
Subsidiary R&D	228	0.45	0.49	0	1
Subsidiary autonomy	228	0.58	0.49	0	1
% of managers with MNE experience	228	16.30	28.01	0	100
Subsidiary CEO industry experience	228	14.64	10.51	1	50
Average age of managers	228	38.00	5.80	23	58
% of managers with MBA degree	228	16.26	27.22	0	100

4.2 | Effect of subsidiary managers' characteristics on subsidiary innovation

Table 5 presents the results of binomial logit estimation to test Hypotheses 1 and 2. Column 1 shows the baseline results when we include only control variables in the model. In this specification, three control variables are statistically significant: high-tech industries ($\beta = 0.913$, $p < .05$), international property rights protection ($\beta = 0.509$; $p < .01$), and subsidiary R&D ($\beta = 0.869$; $p < .05$). These results suggest that (a) on average, firms in high-tech industries are more innovative than firms in non-high-tech industries, (b) subsidiaries in countries with stronger property rights protection are more innovative than those in countries with weaker protection of property rights, and (c) subsidiaries that perform R&D are more innovative than those that do not, *ceteris paribus.*

Column 2 indicates the regression results when we include the percentage of subsidiary top managers with MNE prior work experience with the baseline model. The percentage of top managers with prior MNE experience is statistically significant ($\beta = 0.012$; $p < .05$), indicating that subsidiaries with a higher percentage of top managers with MNE work experience are more likely to be innovative. Hypothesis 1 is supported. Column 3 presents the regression results when we add the subsidiary CEO's industry experience to the baseline model. The regression result shows that the CEO's industry experience is statistically significant ($\beta = 0.050$; $p < .01$), indicating that subsidiaries where the top manager has more industry experience engage in more innovation. Hypothesis 2 is supported. In column 4, we include all the hypothesized effect and find similar results as reported earlier.

4.3 | Moderating effect of subsidiary autonomy and R&D activity

Table 6 presents the results of our moderating effects. Column 1 in Table 6 presents the regression results when the interaction of subsidiary autonomy with managers' prior MNE work experience and with the subsidiary CEO's industry experience are added to model 4 in Table 5. The interaction between the percentage of subsidiary managers with prior MNE work experience and subsidiary autonomy is negative and significant ($\beta = -0.017$; $p < .05$), while the direct effect of managers' international experience remains positive and significant ($\beta = 0.023$; $p < .05$). These results support hypothesis 3a, suggesting that subsidiary managers' MNE work experience has a smaller impact on subsidiary innovation when the subsidiary is autonomous. The interaction between the subsidiary

TABLE 3 Correlation matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Product innovation (1)	1.00											
Subsidiary size (2)	0.19	1.00										
Subsidiary age (3)	0.18	0.32	1.00									
Subsidiary patent (4)	0.11	0.16	0.29	1.00								
Size of subsidiary location (5)	0.02	−0.13	0.04	0.07	1.00							
High-tech industries (6)	0.15	0.15	0.14	0.01	−0.02	1.00						
Intellectual property rights protection (7)	0.26	0.28	0.31	0.26	−0.29	0.13	1.00					
Subsidiary R&D (8)	0.24	0.15	0.22	0.29	0.09	−0.01	0.22	1.00				
Subsidiary autonomy (9)	−0.15	−0.13	−0.05	−0.09	0.09	−0.08	−0.27	−0.23	1.00			
% of managers with MNE experience (10)	0.18	0.03	0.06	0.14	−0.12	0.08	0.32	0.16	−0.18	1.00		
Subsidiary CEO industry experience (11)	0.23	0.02	0.16	0.06	−0.01	−0.05	0.19	0.11	−0.02	0.24	1.00	
Average age of managers (12)	−0.04	0.06	0.28	0.11	−0.03	0.08	−0.02	−0.02	0.04	−0.06	0.03	1.00
% of managers with MBA degree (13)	0.01	−0.04	0.05	0.09	−0.06	0.01	0.20	0.09	−0.09	−0.02	0.07	0.06

CEO's industry experience and subsidiary autonomy is positive and significant ($\beta = 0.036$; $p < .05$), while the direct effect of CEO industry experience on subsidiary innovation remains significant ($\beta = 0.029$; $p < .05$). These results indicate that the subsidiary CEO's industry experience has a greater effect on subsidiary innovation when the subsidiary is autonomous, providing support for Hypothesis 3b.

Column 2 in Table 6 presents the regression results when we add the interaction of the subsidiary's R&D with the percentage of subsidiary managers with prior MNE work experience and with the subsidiary CEO's industry experience in model 4 in Table 5. The interaction between the percentage of subsidiary managers with prior MNE work experience and the subsidiary's R&D is negative and significant ($\beta = -0.019$; $p < .05$), while the direct effect of managers' international experience remains positive and significant ($\beta = 0.021$; $p < .05$). These results support Hypothesis 4a, signifying that subsidiary managers' prior MNE work experience has a smaller impact on subsidiary innovation when the subsidiary is active in R&D. The interaction between the subsidiary CEO's industry experience and subsidiary R&D is not statistically significant, despite its positive coefficient ($\beta = 0.014$; $p > .10$). Hypothesis 4b is not supported. The direct effect of the subsidiary CEO's industry experience on subsidiary innovation remains significant ($\beta = 0.046$; $p < .01$). In the last model, we include all interaction variables in a single equation. In general, we find that our results in the last model are consistent with prior models presented in columns 1 and 2.

4.4 | Robustness check

To check the robustness of our results, we run two alternative methods, a two-stage probit estimation and the Heckman selection model. We use the two-stage probit estimation to deal with the potential endogeneity problem caused by variable autonomy, which may be related to many factors (including managerial characteristics and the innovation mandate from headquarters) that eventually are associated with subsidiary innovation. Similarly, we use the Heckman selection model to deal with endogeneity caused by sample selection.

For the two-stage probit estimation, we first run a probit estimation of subsidiary autonomy on the number of staff members reporting directly to the subsidiary CEO, the ratio of nonproduction workers to production workers, subsidiary size, subsidiary age, patent, dummy of high-tech industries,

TABLE 4 Cross-tabulation and association between binary variables

		Subsidiary R&D		Subsidiary autonomy		Subsidiary patent	
		0	1	0	1	0	1
Product innovation	0	59 (25.88%)	10 (4.39%)	21 (9.21%)	48 (21.05%)	48 (21.05%)	21 (9.21%)
	1	75 (32.89%)	84 (36.84%)	75 (32.89%)	84 (36.84%)	95 (41.67%)	64 (28.07%)
	Cramer's V	0.3578		−0.1557		0.0933	
Subsidiary R&D	0			43 (18.86%)	91 (39.91%)	98 (42.98%)	36 (15.79%)
	1			53 (23.25%)	41 (17.98%)	45 (19.74%)	49 (21.49%)
	Cramer's V			−0.2422		0.2572	
Subsidiary autonomy	0					55 (24.12%)	41 (17.98%)
	1					88 (38.60%)	44 (19.30)
	Cramer's V					−0.0957	

international property rights protection score, and subsidiary R&D. The first two independent variables—the number of staff members reporting directly to the subsidiary CEO and the ratio of non-production workers to production workers—serve as instrumental variables because they do not have a direct influence on subsidiary innovation and for this reason, they do not appear in the second-stage estimation. In the second-stage regression, we run subsidiary innovation (measured as product introduction) on our main explanatory variables, moderating variables, and control variables. In the second-stage regression, we use the predicted value of subsidiary autonomy from the first regression to replace the actual value of subsidiary autonomy. A majority of results from the two-stage probit estimation remain qualitatively similar to prior estimations presented in Tables 5 and 6.

We use the result of the first-stage probit estimation as explained above as the basis to calculate the inverse Mills ratio. In the case of the Heckman model, the number of staff members reporting directly to the subsidiary CEO and the ratio of nonproduction workers to production workers serve

TABLE 5 Effect of subsidiary-level and individual-level factors on subsidiary innovation (H1, H2)

	(1)		(2)		(3)		(4)	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Subsidiary size	0.214	(0.136)	0.245*	(0.143)	0.265**	(0.131)	0.298**	(0.143)
Subsidiary age	0.216	(0.146)	0.249	(0.165)	0.174	(0.182)	0.199	(0.208)
Subsidiary patent	−0.084	(0.274)	−0.114	(0.264)	−0.076	(0.283)	−0.100	(0.264)
Size of subsidiary location	0.126	(0.111)	0.133	(0.109)	0.122	(0.117)	0.131	(0.116)
High-tech industries	0.913**	(0.402)	0.880**	(0.374)	0.938**	(0.392)	0.917***	(0.382)
Int'l property rights protection	0.509***	(0.072)	0.443***	(0.076)	0.441***	(0.077)	0.371***	(0.076)
Average age of managers	−0.022	(0.022)	−0.021	(0.021)	−0.029	(0.023)	−0.028	(0.023)
% of managers with MBA degree	−0.005	(0.005)	−0.006	(0.006)	−0.005	(0.006)	−0.007	(0.006)
Subsidiary R&D	0.869**	(0.531)	0.803**	(0.374)	0.832**	(0.398)	0.778***	(0.386)
Subsidiary autonomy	−0.216	(0.370)	−0.177	(0.386)	−0.215	(0.369)	−0.176	(0.377)
% managers with MNE experience			0.012**	(0.006)			0.012**	(0.006)
Subsidiary CEO industry experience					0.050***	(0.009)	0.050***	(0.009)
Number of observations	228		228		228		228	
Log pseudo-likelihood	−122.69		−121.07		−118.27		−116.58	
Pseudo R-square	13.24%		14.39%		16.37%		17.56%	

Notes. Dependent variable equals "1" if firm launches new product or significantly modifies existing product and "0" otherwise. Robust standard errors are in parentheses. ***, **, and * denote 99, 95, and 90% levels of significance, respectively. We use clustered standard errors, with country as the basis for clustering.

TABLE 6 Moderating effect of subsidiary autonomy and subsidiary R&D on the relationship between managerial characteristics and subsidiary innovation (H3, H4)

	(1)		(2)		(3)	
	Coeff.	S.E.	Coeff.	Std. error	Coeff.	Std. error
Subsidiary size	0.296**	(0.135)	0.319**	(0.141)	0.323**	(0.141)
Subsidiary age	0.192	(0.202)	0.169	(0.204)	0.142	(0.206)
Subsidiary patent	−0.135	(0.277)	−0.043	(0.274)	−0.054	(0.284)
Size of subsidiary location	0.118	(0.120)	0.141	(0.122)	0.129	(0.126)
High-tech industries	0.908**	(0.399)	0.929**	(0.393)	0.918**	(0.411)
Int'l property rights protection	0.380***	(0.078)	0.349***	(0.078)	0.353***	(0.081)
Average age of managers	−0.026	(0.022)	−0.027	(0.021)	−0.024	(0.022)
% of managers with MBA degree	−0.008	(0.005)	−0.008	(0.006)	−0.009	(0.005)
Subsidiary R&D	0.772**	(0.375)	0.847*	(0.461)	0.906*	(0.469)
Subsidiary autonomy	−0.434	(0.585)	−0.179	(0.367)	−0.315	(0.353)
% managers with MNE experience	0.023***	(0.007)	0.021**	(0.009)	0.034***	(0.011)
Subs. CEO industry experience	0.029*	(0.016)	0.046***	(0.013)	0.031*	(0.018)
% managers with MNE experience × subs. autonomy	−0.017**	(0.007)			−0.019**	(0.008)
Subs. CEO industry experience × subsidiary autonomy	0.036**	(0.017)			0.031*	(0.018)
% Managers with MNE experience × sub. R&D			−0.019**	(0.008)	−0.021***	(0.011)
Subs. CEO industry experience × subsidiary R&D			0.014	(0.019)	0.011	(0.019)
Number of observations	228		228		228	
Log pseudo-likelihood	−115.41		−115.63		−114.38	
Pseudo R-square	18.39%		18.23%		19.11%	

Notes. Dependent variable equals "1" if firm launches new product or significantly modifies existing product and "0" otherwise. Robust standard errors are in parentheses. ***, **, and * denote 99, 95, and 90% level of significance, respectively. We use clustered standard errors, with country as the basis for clustering.

as exclusion restrictions. We then run a probit regression of subsidiary innovation on our main explanatory variables, moderating variables, control variables, and inverse Mills ratio. Again, a majority of results from the Heckman selection model remain qualitatively similar to prior estimations presented in Tables 5 and 6. The inverse Mills ratio in this model is not statistically significant, which indicates that our original logit estimation may not suffer from selection bias.

4.5 | Graph simulation

We further illustrate how subsidiary autonomy and R&D activity moderate the effect of subsidiary managers' MNE work experience and the CEO's industry experience on the propensity to innovate by creating a simulation using the binomial logit regression results in Table 5, columns 1 and 2. The predicted probability to innovate follows a variant of the cumulative logistic function. Figure 2 illustrates the moderating effect of subsidiary autonomy and R&D activity on the relationship between subsidiary managers' MNE work experience and innovation, where the first panel (on the left) separates the autonomous and nonautonomous subsidiaries and the second panel separates subsidiaries with and without R&D activities. In this simulation, we include only statistically significant variables from columns 1 and 2 of Table 6, and the percentage of subsidiary managers with prior MNE work experiences ranges from 0 to 100%. Furthermore, we assign the mean value to replace the remaining significant variables included in the simulation (see Table 2 for the mean value of each variable), and by doing so, we assume that the subsidiary has an average value for all characteristics we capture in our model. The graph in the first panel shows that as the percentage of managers with prior MNE work experience goes up, the predicted probability of a nonautonomous subsidiary (light gray line) increases at a faster rate than that of an autonomous subsidiary (dark gray line). It is also noteworthy that the predicted probability of innovation in a nonautonomous subsidiary is smaller than that of an autonomous subsidiary when the percentage of managers with MNE work experience is less than 30%. This is due to the greater marginal effect of industry experience in an autonomous than in a nonautonomous subsidiary. The second panel of Figure 2 shows the moderating effect of subsidiary R&D. As the percentage of managers with MNE experience rises, a subsidiary without R&D (dark gray line) enjoys a faster rate of change in predicted probability compared to a subsidiary with R&D.

Figure 3 illustrates the moderating effect of subsidiary autonomy on the relationship between the subsidiary CEO's industry experience and innovation. The graph shows that the predicted probability of innovation by autonomous subsidiaries (dark gray line) increases at a faster rate than that of nonautonomous subsidiaries as the years of the subsidiary CEO's industry experience increase.

5 | DISCUSSION AND CONCLUSION

This study attempts to incorporate the emergent microfoundations approach to strategy and prior works on headquarters–subsidiary relationships to explain innovation at the subsidiary level. The extant literature on microfoundations of global strategy (Felin et al., 2015; Morris, Hammond, & Snell, 2014) focuses on the characteristics of top managers at the MNE level and, therefore, little is known about the effect of managerial characteristics at the subsidiary level. We argue that subsidiary managers are important boundary spanners, integrators, and connectors scouting for knowledge and capabilities from within the MNE as well as from the external environment which, in turn, influences subsidiary innovation. The internal and external boundary-spanning capabilities of subsidiary managers are reflected in MNE-specific experience and industry experience, respectively. We further argue that the internal and external boundary-spanning capabilities have differential effects on innovation depending on the level of a subsidiary's autonomy and its engagement in R&D activities.

We tested our hypotheses on a sample of MNE subsidiaries in 11 countries by utilizing a unique dataset available in World Bank Enterprise surveys. Our logit regression results show that subsidiaries with a greater percentage of managers with prior MNE work experience or longer industry experience are more likely to innovate. These results are robust after controlling for various managerial-, firm-, industry-, and country-level characteristics. We also find that subsidiary autonomy negatively moderates the effect of managers' prior MNE work experience and positively moderates the effect of managers' industry experience on innovation. We also find evidence that a subsidiary's R&D positively moderates the relationship between managers' industry experience and innovation.

Our study has important implications for MNEs' global strategy. First, our study suggests that staffing strategies for subsidiaries are relevant for MNEs' global competitiveness. Managers with prior MNE work experience and managers with longer industry-specific experience help enhance subsidiary innovation. Second, the suggested effects of managers' experiences are not context-free, but vary depending on the trade-off between global integration versus local adaptation and parent knowledge versus local knowledge. For MNEs that emphasize global integration and rely on parent knowledge, managers with MNE work experience should be given higher priority. However, for

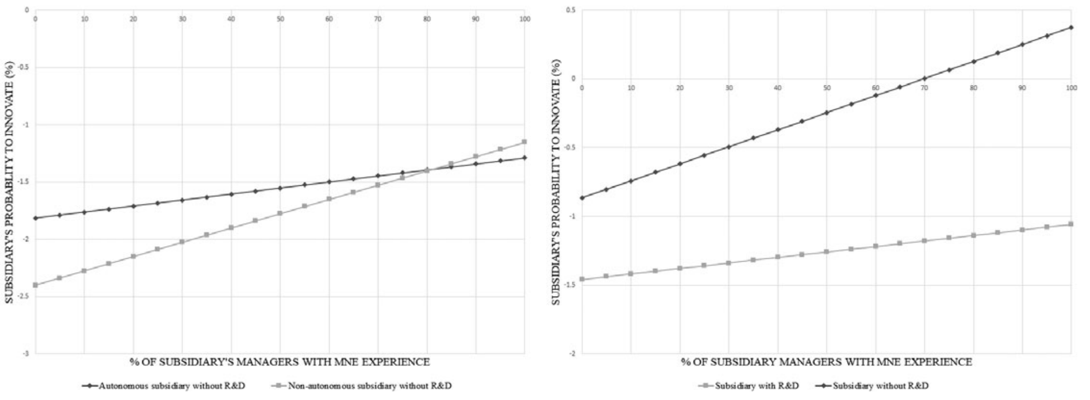

FIGURE 2 Moderating effect of subsidiary autonomy and R&D on managers' MNE work experience and innovation. *Notes:* Predicted probability is calculated based on coefficients presented in columns 1 and 2 in Table 6. We include only significant variables and assign mean values to those included variables (except percentage of subsidiary's managers with MNE work experience)

MNEs that pursue local adaptation, industry experience is more critical than prior MNE work experience.

We make several contributions to the literature on knowledge management in global organizations. First, our study is a response to recent calls to examine the microfoundations approach to knowledge transfer mechanisms, one of the thorniest challenges facing the modern MNE (Andersson et al., 2016; Cano-Kollman et al., 2016). We incorporate this microfoundations approach by studying how subsidiary managers' experiences affect the delicate balance between headquarters and subsidiaries in the knowledge generation process. Our study highlights the important role that key individuals play in the knowledge development process, thus contributing to the microfoundations and innovation literatures.

Second, the use of new product development as the dependent variable is a refined indicator of innovation performance because it represents actual output rather than the ubiquitous R&D intensity, patents, and citations, which are input and intermediate indicators (Hagedorn & Cloodt, 2003). Thus, our results carry more explanatory power because we examine the link between the subsidiary managers' characteristics to tangible knowledge generation outcomes. Third, our theory proposes subsidiary managers as boundary spanners, integrators, and connectors of knowledge development more than just in terms of interpersonal relationships (Johnson & Duxbury, 2010), trust and cultural isolation (Raab, Ambos, & Tallman, 2014). Fourth, the results of our study show the moderating effects of a subsidiary's autonomy and a subsidiary's R&D on the relationship between boundary-spanning characteristics such as MNE experience and industry experience and new product development in the MNE subsidiaries.

Our study also has important managerial implications. In an extremely competitive global marketplace, MNEs must keep innovating to maintain their competitive advantage. One of the ways to do this is by making sure that more subsidiary managers have both MNE experience and industry experience. MNEs need subsidiary managers to act as boundary spanners so they can access knowledge inside the MNE and outside the MNE as connectors to distant and diverse sources of knowledge. This requires recruitment of managers with the aforementioned experience as well as appropriate compensation and retention plans.

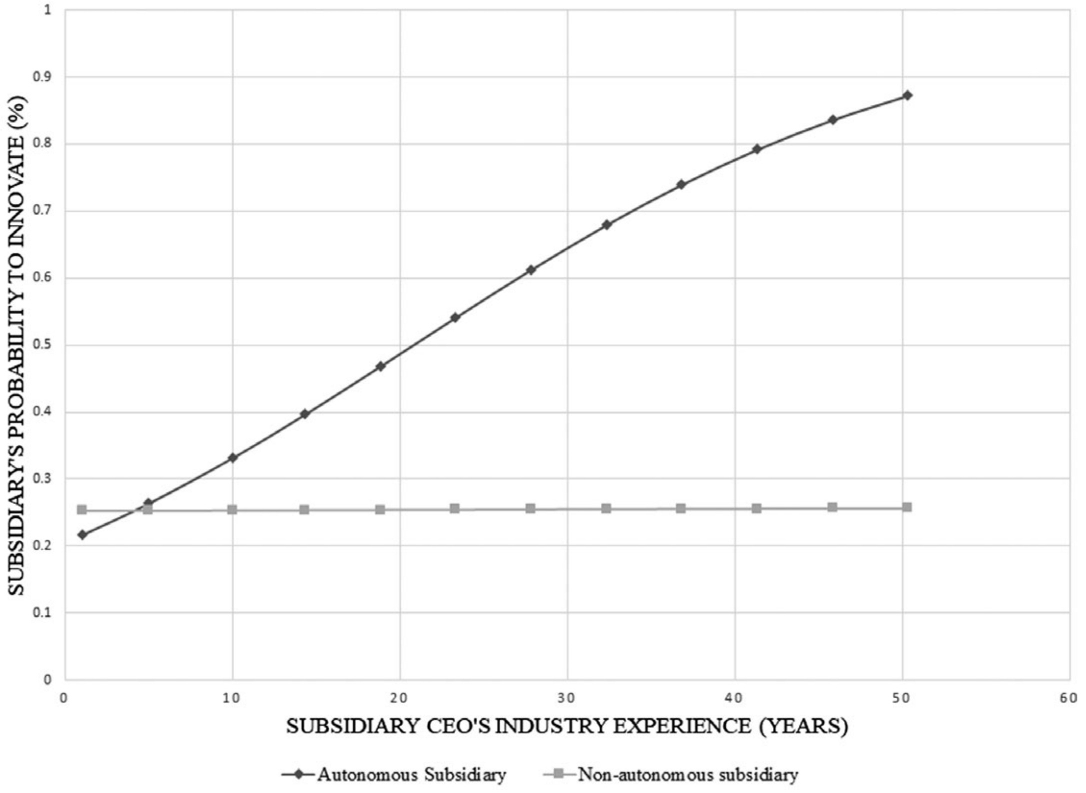

FIGURE 3 Moderating effect of subsidiary autonomy on subsidiary CEO's industry experience and innovation.
Notes: Predicted probability is calculated based on coefficients presented in column 1 in Table 6. We include only significant variables and assign mean values to those included variables (except subsidiary CEO's industry experience, which follows the range from 1 to 50 years)

Our research is not without limitations. Because the sample consists of 11 countries, and some of the countries include small samples (India and Russia), caution is needed to make broad generalizations. Managerial characteristics by their nature are a proxy for executives' behaviors and may not reflect cognitive behavior closely. The sample is somewhat limited because we used the data collected by the World Bank. Additionally, new product introduction as a measure of subsidiary innovation has limitations because we are unable to assess basic innovation, which may be reflected in visible output. Notwithstanding these data limitations, we do theoretically examine the underlying mechanisms through which subsidiary managers affect subsidiary innovation. Future research should incorporate demographic characteristics such as functional background, firm tenure, socioeconomic status, and cognitive schema of subsidiary managers as determinants of subsidiary-level outcomes. Additionally, researchers need to incorporate previous knowledge flows, because that might affect current new product development. Finally, cross-sectional and time-varying data may shed more light on the subsidiary knowledge generation process.

In conclusion, we explain subsidiary innovation based on the subsidiary managers' internal and external boundary spanning capabilities, the effects of which vary depending on the nature of the MNC-subsidiary relationship and the focus on local innovation in the subsidiary. We hope that this research encourages scholars to examine further the organizational processes that make some global organizations more innovative than others.

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